



University  
of Glasgow

December 2023

DEGREES of MSci, MEng, BEng, BSc, MA and MA (Social Sciences)

# **Introduction to Data Science and Systems COMPSCI 5089**

(Answer All 4 Questions)

**This examination paper is an open book, online assessment and  
is worth a total of 80 marks.**

1. **Note: For this question, you can write Maths in a Numpy ndarray style to facilitate your typing, e.g.**

$$C = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$$

**can be written by  $C = [[3, 1], [2, 2]]$ .**

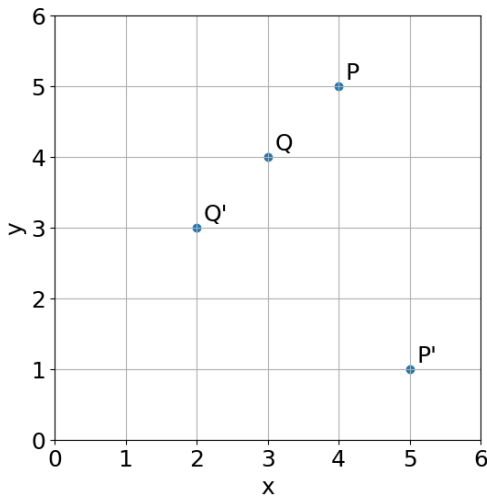
- (a) For a project, your data is contained in a matrix  $M$  of 100 rows and 10 columns, where each row represents a different subject, and each column represents a recorded observation. Explain all steps required for whitening this dataset, explaining the dimension of all matrices and vectors involved.

[6]

- (b) Assume that two of your variables are H: the subject's height (in meters) and W: the subject's weight (in pounds). What would be the effect of whitening on those two variables and why (and when) would it be desirable? Give an example.

[4]

- (c) Assume the linear function described by the matrix  $M$  maps the two points  $P \rightarrow P'$  and  $Q \rightarrow Q'$ , as in the figure below:



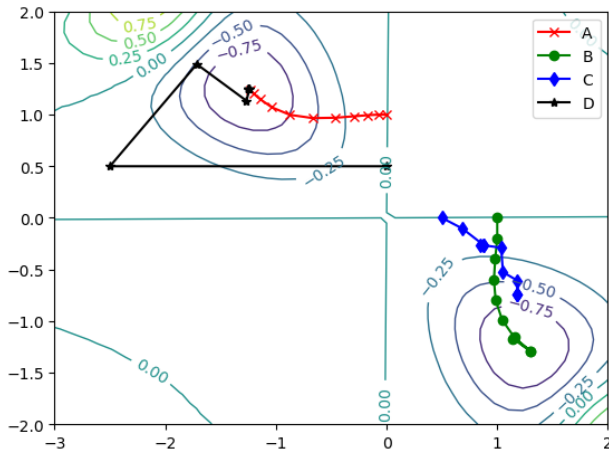
such that  $P' = MP$  and  $Q' = MQ$ .

- (i) Explain how you would calculate the matrix  $M$  from the coordinates of those two point correspondences *using linear algebra*. Explain all the steps of your reasoning and provide all relevant equations and matrices that you would use for this calculation. You do not need to calculate the actual values, just to provide the method you would use (for example to solve it with numpy).
- (ii) Imagine you had a third point correspondence  $R \rightarrow R'$ . Could you use the same approach, and if not how could you find  $M$ ?

[7]

[3]

2. (a) Look carefully at the plot in the image below.



This is a contour plot of the loss function you are trying to minimise.

Explain how you would best qualify the loss function at the following points, and what you would expect the gradient to be and why:

- at  $(-1, -1)$
- at  $(-1, 1)$
- at  $(-2, 1.5)$

(comment on both the direction and magnitude of the gradient)

[6]

(b) The red (A), green (B), blue (C) and black (D) lines show the trajectories of four iterative optimisation algorithms that a colleague of yours attempted (each had a different starting point).

Looking at the graphs:

- Which one of the three is most likely to be a *gradient descent* algorithm?
- Which one is most likely to be a *hill-climbing* algorithm?
- Which one is most likely to be a *Newton method*?

Explain why in each case.

[6]

(c) What could be an issue when optimising this loss with iterative algorithms (for example gradient descent)? What solution could you suggest to mitigate this issue? Looking at the graph, do you think this is likely to be a problem for your application?

[5]

(d) Assume that you want your optimisation to *not* find the (local) minimum to the bottom right of the graph. Explain in details how to achieve this.

[3]

3. Assume that you are looking at students' performance on a specific university module. You have collected examination results for a small cohort. You have the marks for each student for the exam and the coursework (out of 10), as well as the age and study group (A or B) of the students.

The data is collated in the following data table:

|   | age | exam | coursework | group |
|---|-----|------|------------|-------|
| 0 | 21  | 9    | 7          | A     |
| 1 | 18  | 3    | 7          | A     |
| 2 | 18  | 5    | 6          | A     |
| 3 | 19  | 6    | 8          | B     |
| 4 | 18  | 4    | 5          | B     |

**Note: For this whole question, answers in python code or pseudo code are acceptable alternative to classical mathematical formulation.**

- (a) Calculate the *expectation* for the exam mark and for the coursework mark.

You are told that the final module mark is calculated from the exam and coursework by applying a weighting of 60% to the exam score and of 40% to the coursework score. Calculate the expectation of the final mark for this cohort.

[6]

- (b) Assume that the pass grade for the module is 5 out of 10 (ie, students with a mark lower than 5 would fail and students with 5 or above would pass).

- From the data above, calculate an estimate of the probability of getting a passing mark.
- The probability of getting a passing mark for students in group A and in group B.
- What can you conclude from this?

[7]

- (c) Now assume that you have three pieces of coursework and one exam, and data at your disposal for large student cohorts over several years. You now want *to set the weighting of each mark* to approach a pass rate of 80%. Describe the approach you would follow to achieve this (you are not expected to do the actual calculations).

[7]

4. Consider the following two relations:

- Course(Id, Description, Credits), or  $C$ , where:
  - Id, the primary key of this relation, is a 32-bit integer,
  - Description is a 195-byte string, and
  - Credits is an 8-bit integer.
- Transcript(StudentId, CourseId, Mark), or  $T$ , where:
  - StudentId is a foreign key that has 4 byte.
  - CourseId is a foreign key to the primary key (Id) in the Course relation above
  - Mark is a 8-byte double precision floating number, and
  - the primary key consists of the combination of StudentID and CourseID.

Relation Course ( $C$ ) has  $r_C = 32$  records, Relation Transcript ( $T$ ) has  $r_T = 51200$  records. Given the relations  $C$  and  $T$  described previously:

- (a) Calculate the blocking factor (number of records per block) for both relations  $C$  and  $T$ . [4]
- (b) Compute the number of blocks required to store the relations  $C$  and  $T$  based on their blocking factors. [6]
- (c) Using a nested loop join, derive the cost analysis for joining these two relations. Namely, compare the cost for the strategies when:
- Course  $C$  is used as the outer relation and Transcript  $T$  is used as the inner relation.
  - Transcript  $T$  is used as the outer relation and Course  $C$  is used as the inner relation.
- Given that the memory of the database system can accommodate (  $n_B = 23$  ) blocks for processing and considering (  $n_B - 2$  ) blocks for reading the outer file, find which strategy is more efficient. [6]
- (d) Explain the concept of “Selection Cardinality” and briefly describe its significance in query optimisation. [4]